Zafaryab Haider

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SUMMARY

AI security and trustworthy-ML researcher: adversarial and bit-flip attacks on multimodal and speech models, RLHF training-signal integrity, harm evaluation for multi-agent LLM systems, and the evaluation tooling behind them.

EXPERIENCE

University of Maine | Graduate Research/Teaching Appointments | Orono, ME ***August 2022 – Present***

- Built **BLADE** and **SMASH**, differentiable bit-level fault-injection frameworks for quantized vision–language and speech-recognition models; BLADE reached penalized ASR **48.80 vs. 20.95** (random-proposal baseline, 20-bit budget).
- Built **COBRA** and **ACT**: consensus-based reward for RLHF under malicious feedback (∼**73% vs. 48.78%**, ∼**23%** over prior work at DF=10) and anchor-calibrated trusted/corrupted cohort recovery.
- Built distributed LLM inference across **heterogeneous NVIDIA Jetson and RTX GPUs**; models up to 15B.
- **AI/ML Consultant**, ARCSIM (2024–2026); guest lectures for **ECE 479/579** and the UMaine **AI Seminar**.

Aligarh Muslim University | Assistant Professor | Aligarh, India ***January 2015 – August 2022***

- Designed and taught information security, networking, data structures and C programming to **200+ students annually**.
- Led and mentored applied ML and systems projects across healthcare ML, IoT sensing and intelligent decision support.
- Created an open-access YouTube channel for remote course material during COVID-19; one video reached 116K+ views.

SELECTED RESEARCH & IP

Accepted, published & under review

- **Haider, Z.** et al. *SMASH: Probing Speech Recognition Robustness via Semantically Targeted Bit Flips.* **NeurIPS**.
- Rahaman, Xu, **Haider, Z.** et al. *(A)iSpy: Parasitic Trojans for Machine Learning Infrastructure.* **ACM CCS**.
- **Haider, Z.** et al. *ACT: Anchor-Calibrated Trust Estimation When Corrupted Sources Form Majority.* **IEEE CARS'26**.
- Nafi, **Haider, Z.** et al. *DASH: Meta-Attack Framework for Effective & Stealthy Adversarial Examples.* **CVPR**.
- **Haider, Z.** et al. *A Framework for Mitigating Malicious RLHF Feedback in LLM Training using Consensus Based Reward.* **Nature Scientific Reports**, 2025.
- **Haider, Z.** et al. *BLADE: Bit-Flip Attacks for Semantic Misdirection in Quantized Image Captioning.* Under review.

Patents & IP — five invention families

- **Automated Neural Architecture Generation.** PCT Application No. PCT/US2025/013847, 2025.
- **COBRA.** PCT Application No. PCT/US26/18361, 2026.
- **BLADE.** U.S. Provisional Patent Application No. 63/930,131, 2025.
- **Explainable Agentic Cross-PDK Migration Difficulty.** U.S. Provisional Patent Application No. 64/075,062, 2026.
- **HARP: Multi-Agent LLM Harm Amplification.** U.S. Provisional Patent Application No. 64/148,848, 2026.

SELECTED PROJECTS & SYSTEMS

REWIRE: Grid Reconfiguration World Models August 2026 – Present
Topology-aware world model ranking grid line-switching candidates via learned rollouts, verified by exact AC power flow.

HARP — Multi-Agent LLM Harm Amplification September 2025 – October 2026
Benchmark and methodology measuring how compromised agents amplify local failures into system-level harm.

Edge-AI Research Testbed — 45 compute devices: 21 Jetsons, 10 Raspberry Pis, 14 SBCs University of Maine
Planned and led its two-rack deployment (remote SSH, shared NAS) with a graduate researcher; guided an undergraduate.

GRANTS & FELLOWSHIPS

SentriAlign: Training-Signal Integrity for LLMs — Co-Investigator & Technical Lead, MIRTA 8.0 ($25,000) 2026
NeurIPS 2026 Financial Assistance Award (full registration and hotel stay) 2026
Maulana Azad National Fellowship (MANF), Government of India 2014
Graduate Aptitude Test in Engineering (GATE) Scholarship, Government of India August 2011 – July 2013

TECHNICAL SKILLS

Languages & Tools: Python, C/C++, SQL, Shell, Git, Jupyter, LaTeX
Frameworks: PyTorch, scikit-learn, NumPy, CuPy, Pandas, Streamlit, REST APIs
AI/ML & Security: adversarial ML, bit-flip & fault injection, RLHF and reward modeling, LLM and agentic evaluation
Systems & Edge: Slurm, Apptainer, NVIDIA containers, multi-GPU/HPC, GCP, Jetson (Orin Nano, Xavier NX, AGX)

EDUCATION

Ph.D., Electrical and Computer Engineering — University of Maine, Orono, ME, USA Expected March 2027
M.Tech., Computer Engineering — Aligarh Muslim University, India (GPA 9.33/10.0) August 2011 – July 2013